# Project Design Phase

## Solution Architecture

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| Date | 15 Febuary 2026 |
| Team ID | GACSE-114 |
| Project Name | FitBuddy: AI Fitness Plan Generator Using Gemini Model |
| Maximum Marks | 4 Marks |

# Solution Architecture Description

FitBuddy AI utilizes a web-based AI application architecture designed to bridge the gap between complex health data and user-friendly fitness guidance. The system is built on a **Streamlit-based** framework that manages the end-to-end flow from user input to personalized health output.

### Core Functional Goals:

- **Problem Solving:** Provides a technical solution for users seeking personalized fitness and nutrition guidance.
- **Structural Definition:** Defines the behavior of the AI model and backend calculations to project stakeholders.
- **Requirement Delivery:** Manages specific health requirements like BMI and TDEE calculations to deliver actionable plans.

2. System Components & Workflow

The architecture is divided into distinct processing layers to ensure scalability and real-time guidance:

- **Frontend (Streamlit UI):** An intuitive web interface where users enter profile data, including age, weight, height, diet preferences, activity levels, and fitness goals.
- **Backend Processing (Python):** This layer acts as the "interaction handler" (similar to a Lambda function). It performs:
    - **Health Metrics:** Calculation of Body Mass Index (BMI) and Total Daily Energy Expenditure (TDEE).
    - **Data Projections:** Utilization of **NumPy** to generate 30-day weight projection charts.

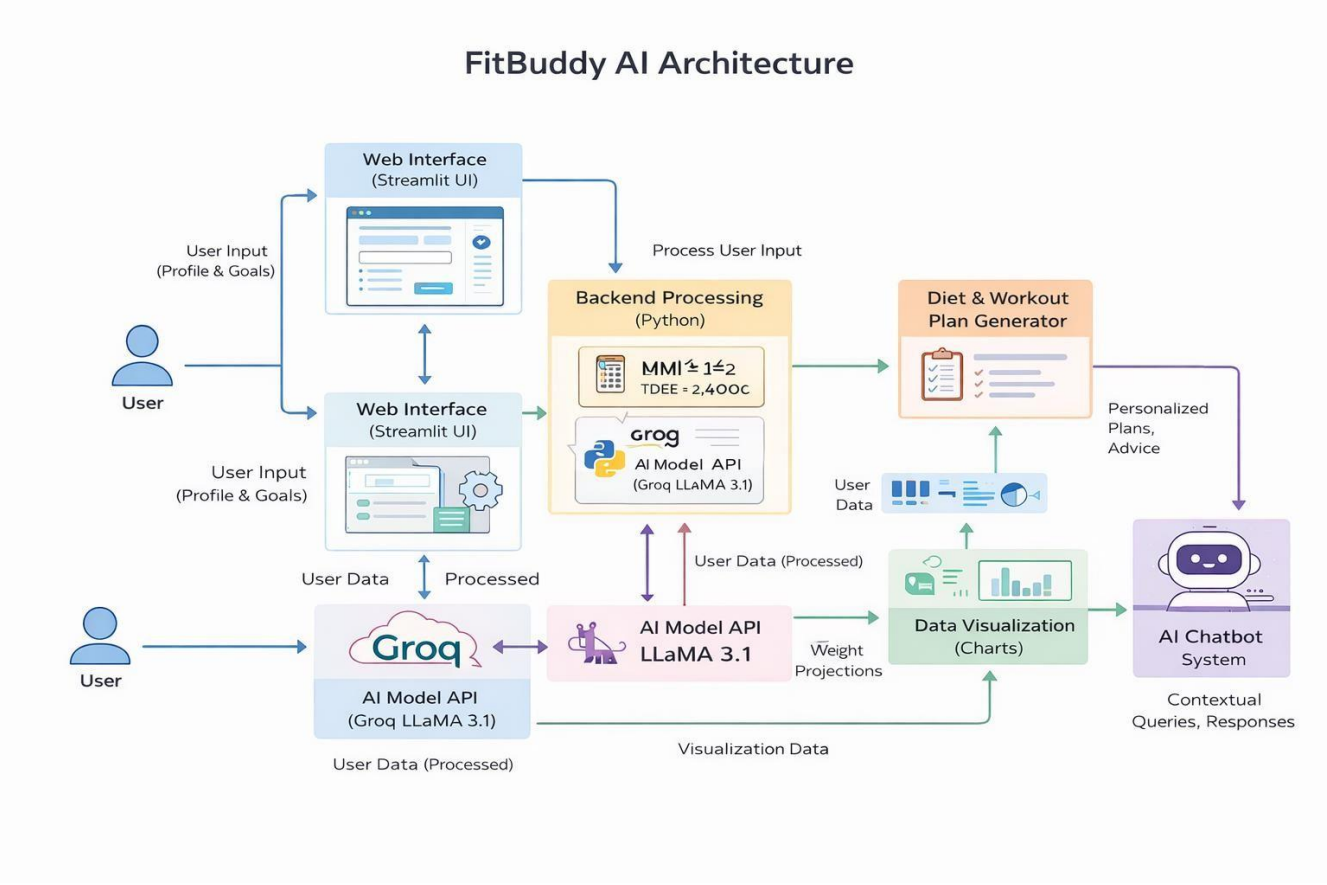
- **AI Integration (Groq API):** The processed user data is sent to the **LLAMA 3.1 8B** language model. This engine generates:
    - Personalized workout routines.
    - Calorie targets and diet plans.
    - Motivational advice.
- **Interactive Chatbot:** A dedicated AI chatbot interface that utilizes **Streamlit session state** to maintain conversation context, allowing users to ask follow-up health questions.

## 3. Data Flow Representation

The data follows a structured path from the user to the final visualization:

1. **Input:** User provides profile and goals through the Web Interface.
2. **Processing:** Python backend calculates core metrics and formats data for the AI.
3. **Inference:** The Groq LLAMA 3.1 model receives processed data to generate plans.
4. **Visualization:** NumPy results and AI outputs are converted into charts and text-based plans.
5. **Feedback Loop:** The AI Chatbot system handles contextual queries and responses for ongoing support.

| **Component** | **Technology** |
|---|---|
| **Frontend/UI** | Streamlit |
| **Programming** | Python / NumPy |
| **AI Model** | LLAMA 3.1 8B (via Groq API) |
| **Health Metrics** | BMI, TDEE |
| **State Management** | Streamlit Session State |


FitBuddy AI Architecture
Web Interface
(Streamlit UI)
User Input
(Profile & Goals)
Process User Input
Backend Processing
(Python)
TDEE = 2,400c
Grog
AI Model API
(Groq LLaMA 3.1)
Diet & Workout
Plan Generator
User
Web Interface
(Streamlit UI)
User Input
(Profile & Goals)
Personalized
Plans,
Advice
User
Data
User Data
Processed
User Data (Processed)
User
Groq
AI Model API
(Groq LLaMA 3.1)
AI Model API
LLaMA 3.1
Weight
Projections
Data Visualization
(Charts)
AI Chatbot
System
Contextual
Queries, Responses
Visualization Data
User Data (Processed)